

- 11.12.** Modify the computer program that was written for problem 11.11 so that it switches to the fourth-order Adams-Bashforth-Moulton numerical integration algorithm after the first three time steps. Thoroughly test your program as was done in problem 11.11.
- 11.13.** Write a computer program to simulate the flight of an airplane with conventional controls (ailerons, elevator, and rudder). Use the fourth-order Runge-Kutta numerical integration algorithm to integrate the Euler-Rodrigues quaternion formulation. Renormalize the quaternion after each full time step by applying Eq. (11.10.7). Use linearized aerodynamics to approximate the forces and moments, but be sure to design the program so that it is easy to change the mathematical model for the forces and moments without altering the rest of the program. Be sure that the inertia tensor is inverted only once, before the simulation is started.
- 11.14.** Use the program that was written for problem 11.13 together with appropriate aerodynamic coefficients and mass properties to repeat the computations made in problem 11.11. Use the following properties for the arrow:

$$\begin{aligned}
 S_r &= 0.15 \text{ ft}^2, \quad l_r = 2.0 \text{ ft}, \quad \rho = 2.0 \times 10^{-3} \text{ slug} \cdot \text{ft}^{-3}, \quad W = 0.0483 \text{ lbf}, \\
 I_{xx_b} &= 1.5 \times 10^{-5} \text{ slug} \cdot \text{ft}^2, \quad I_{yy_b} = I_{zz_b} = 7.5 \times 10^{-4} \text{ slug} \cdot \text{ft}^2, \\
 k_0 &= 0.0061, \quad k_1 = 1.40, \quad k_2 = 0.0059, \quad k_3 = 0.0040, \\
 k_4 &= 0.0080, \quad k_5 = 0.00475, \quad C_{\ell 0} = 0.00495
 \end{aligned}$$

Thoroughly test your program as was done in problem 11.11. Be sure to try several different initial headings.

- 11.15.** Use the program that was written for problem 11.13 together with appropriate aerodynamic coefficients and mass properties to repeat the computations made in Example 10.4.3. Be sure that the airplane is trimmed for the correct airspeed before the step change in elevator deflection is applied. Compare your numerical solution with the analytical solution presented in Example 10.4.3.
- 11.16.** Use the program that was written for problem 11.13 together with appropriate aerodynamic coefficients and mass properties to repeat the computations made in Example 10.4.4. Be sure that the airplane is trimmed for the correct airspeed before the step change in rudder deflection is applied. Compare your numerical solution with the analytical solution presented in Example 10.4.4.
- 11.17.** Modify the computer program written for problem 11.13 so that it switches to the fourth-order Adams-Bashforth-Moulton algorithm after the first three time steps. Also modify the program to renormalize the quaternion only after each 100 time steps. Test your modified program by using it to repeat the computations made in problems 11.14 through 11.16.